

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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Discrete Fourier Transform

- Definition of the Discrete Fourier Transform
- Properties of the Discrete Fourier Transform
- Calculating Linear Convolution using DFT
- Analyzing Signal Spectra using DFT
- Calculating DFT using MATLAB

Properties of the Discrete Fourier Transform

- Linear Properties
- Circular Shift Property
- Symmetry Property
- Circular Convolution Theorem

Properties of the Discrete Fourier Transform

(1) Linear Property

If: $x_1[k] \xleftrightarrow{\text{DFT}} X_1[m]$

$x_2[k] \xleftrightarrow{\text{DFT}} X_2[m]$

Then: $ax_1[k]+bx_2[k] \xleftrightarrow{\text{DFT}} aX_1[m]+bX_2[m]$

If the lengths of two sequences are unequal, the shorter sequence needs to be zero-padded.

$y[k]=x_1[k]+\mathrm{j}x_2[k],\ Y[m]=X_1[m]+\mathrm{j}X_2[m]$

Properties of the Discrete Fourier Transform

(2) Circular Shift Property

Definition of Circular Shift: $x[k] \xrightarrow{Circular} x[(k+n)_N]$

Symbols $(k)_N$: Representing the modulo operation on k by N

If $k = k_1 + k_2 N, \quad k_1 = 0,1,\cdots,N-1, \quad k_2 \in Z$

Then $(k)_N = k_1$ $\quad (k)_N = (k+N)_N = k_1$

Properties of the Discrete Fourier Transform

(2) Circular Shift Property

Definition of Circular Shift: $x[k] \xrightarrow{\text{Cyclic Shift Property}} x[(k+n)_N]$

$k =$ 0, 1, 2, 3, 4, 5, 6, 7, 8

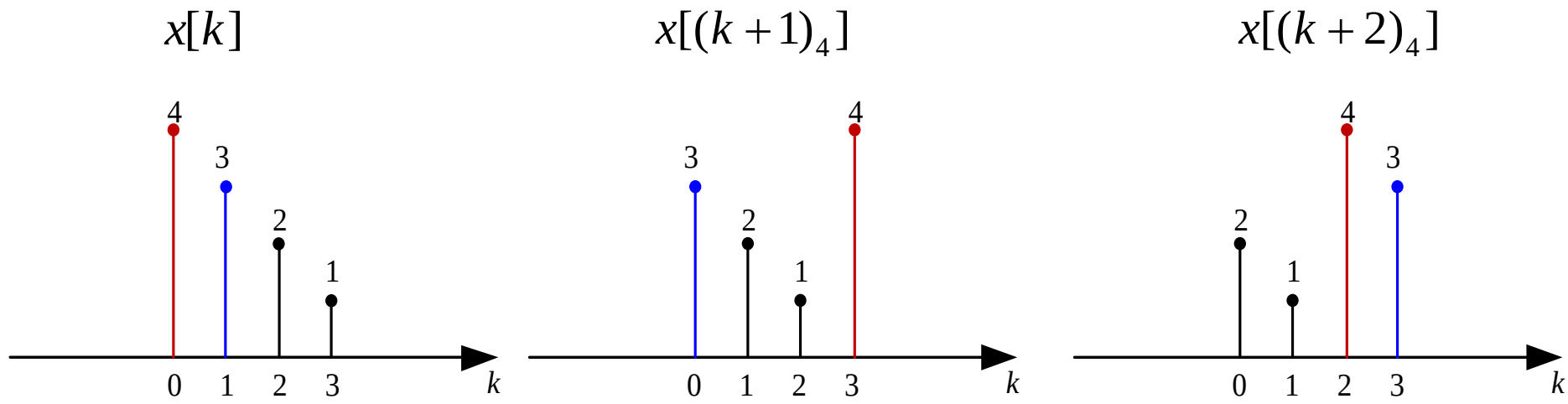
$(k)_4 =$ 0, 1, 2, 3, 0, 1, 2, 3, 0

$k =$ -6, -5, -4, -3, -2, -1, 0, 1

$(k)_4 = (k+8)_4 =$ 2, 3, 0, 1, 2, 3, 0, 1

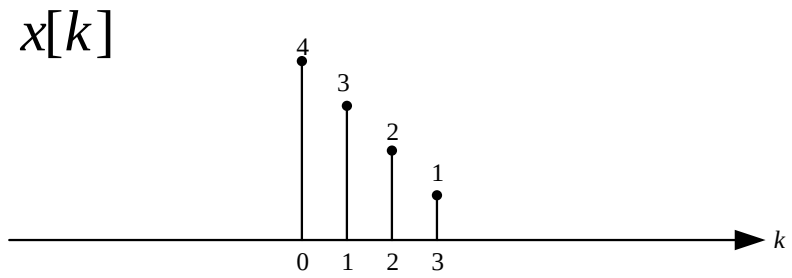
Properties of the Discrete Fourier Transform

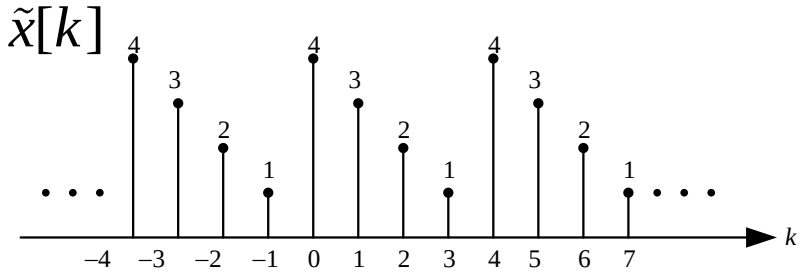
(2) Circular Shift Property

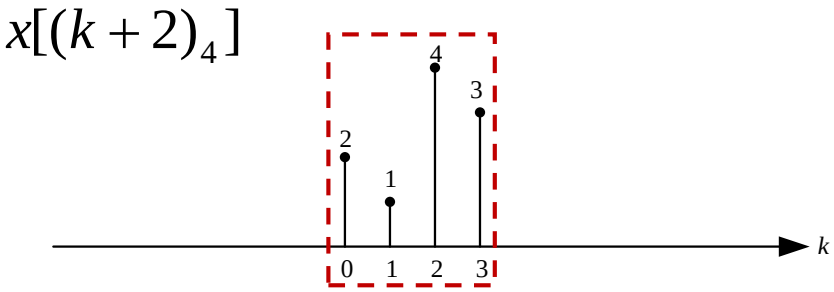


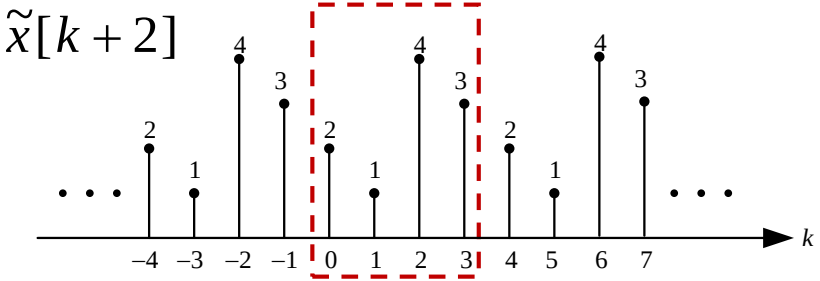
Properties of the Discrete Fourier Transform

(2) Circular Shift Property









Properties of the Discrete Fourier Transform

(2) Circular Shift Property

Time Domain Circular Shift Property

If: $x[k] \xleftrightarrow{\text{DFT}} X[m]$

Then: $x[(k+n)_N] \xleftrightarrow{\text{DFT}} W_N^{-mn} X[m] (\mathrm{e}^{\mathrm{j}\frac{2\pi}{N}mn} X[m])$

Time-domain circular shifts correspond to phase shifts in the frequency domain.

Properties of the Discrete Fourier Transform

(2) Circular Shift Property

Frequency Domain Circular Shift Property

If: $x[k] \xleftrightarrow{\text{DFT}} X[m]$

Then: $W_N^{lk} x[k] (\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}lk} x[k]) \xleftrightarrow{\text{DFT}} X[(m+l)_N]$

Phase shifts in the time domain correspond to circular shifts in the frequency domain.

Properties of the Discrete Fourier Transform

(3) Symmetry Property

- Periodic Conjugate Symmetry

$$x[k] = x^*[(-k)_N] = x^*[(N-k)_N]$$

- Periodic Conjugate Antisymmetry

$$x[k] = -x^*[(-k)_N] = -x^*[(N-k)_N]$$

Properties of the Discrete Fourier Transform

(3) Symmetry Property

※ When the sequence x[k] is a real sequence

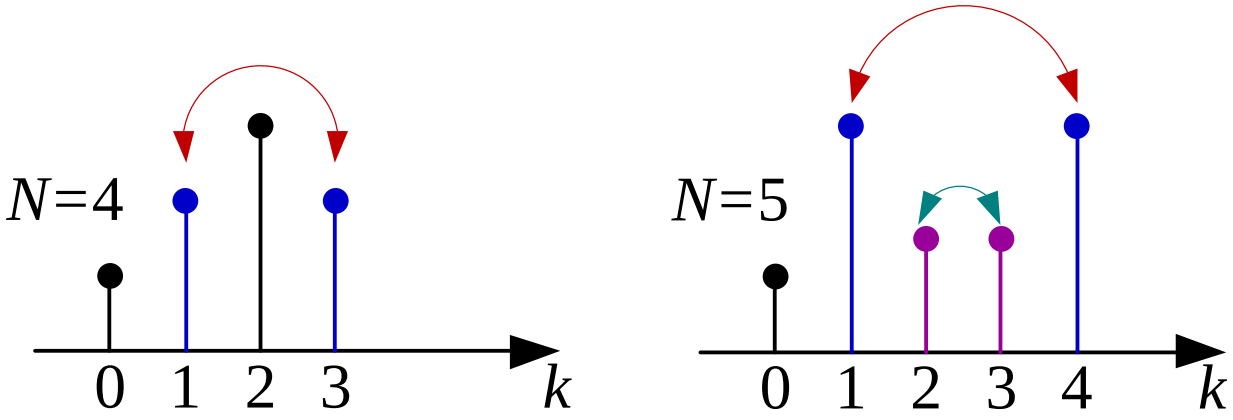
$$x[k] = x[(-k)_N] = x[(N-k)_N]$$ Periodic Even Symmetry

$$x[k] = -x[(-k)_N] = -x[(N-k)_N]$$ Periodic Odd Symmetry

Properties of the Discrete Fourier Transform

※ Real Sequence with Periodic Even Symmetry

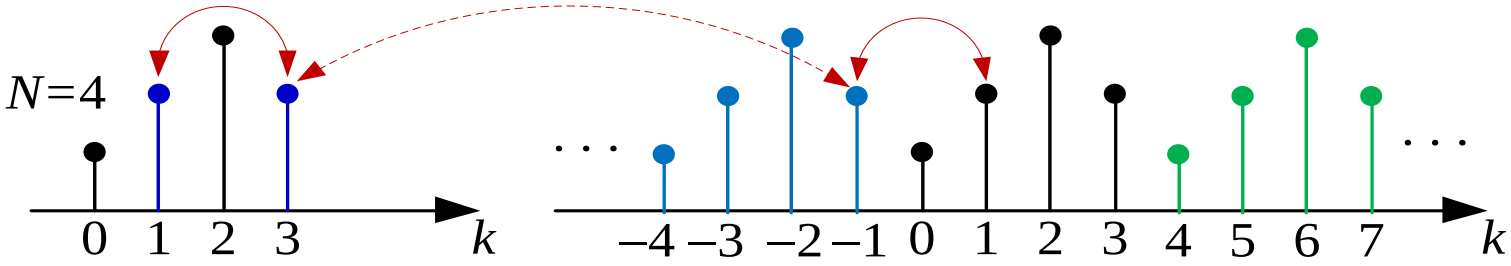
$$x[k] = x[(-k)_N] = x[(N-k)_N]$$



Properties of the Discrete Fourier Transform

※ Real Sequence with Periodic Even Symmetry

$$x[k]=x[(-k)_N]=x[(N-k)_N] \qquad \tilde{x}[k]=\tilde{x}[-k], \quad N=4$$

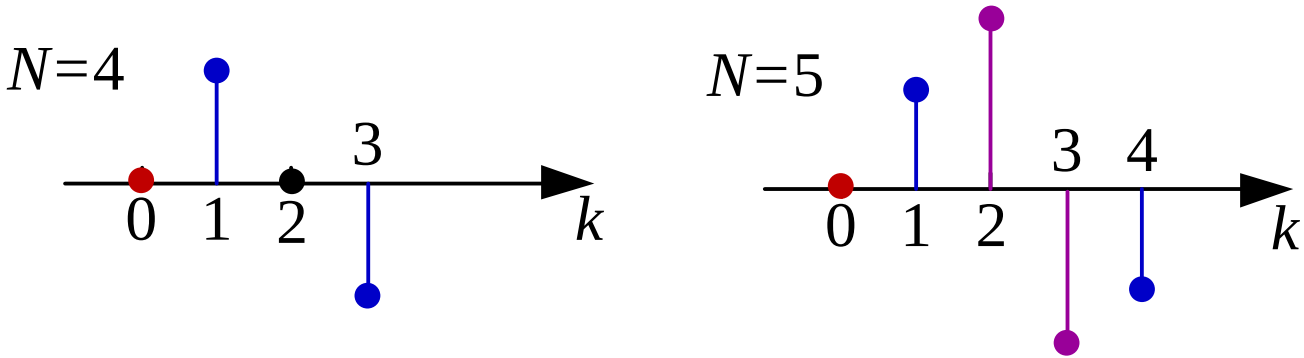


The periodic extension of a finite-length sequence with periodic even symmetry results in an even symmetric sequence.

Properties of the Discrete Fourier Transform

Real Sequence with Periodic Odd Symmetry

$$x[k] = -x[(-k)_N] = -x[(N-k)_N]$$



The periodic extension of a finite-length sequence with periodic odd symmetry results in an odd symmetric sequence.

Properties of the Discrete Fourier Transform

(3) Symmetry Property

If $x[k] \xleftrightarrow{\text{DFT}} X[m]$

Then $x^*[k] \xleftrightarrow{\text{DFT}} X^*[(-m)_N]$

$x^*[(-k)_N] \xleftrightarrow{\text{DFT}} X^*[m]$

Properties of the Discrete Fourier Transform

(3) Symmetry Property

If $x[k]$ is a real sequence, then

$$X[m]=X^*[(-m)_N]=X^*[(N-m)_N]$$

Amplitude and phase

$$|X[m]|=|X[(N-m)_N]|$$

$$\varphi[m]=-\varphi[(N-m)_N]$$

Real part and imaginary part

$$X_{\mathrm{R}}[m]=X_{\mathrm{R}}[(N-m)_N]$$

$$X_{\mathrm{I}}[m]=-X_{\mathrm{I}}[(N-m)_N]$$

Example: Given that the DFT of a certain 9-point real sequence at even-numbered points is $X[0]=3.1$, $X[2]=2.5+4.6j$, $X[4]=-1.7+5.2j$, $X[6]=9.3+6.3j$, $X[8]=5.5-8.0j$。Determine the DFT values at the odd-numbered points.。

Solution: According to the symmetry property of the DFT of a real sequence

$$X[m]=X^*[(-m)_N]=X^*[(N-m)_N]$$

It can be obtained that

$$X[1]=X^*[(9-1)_9]=X^*[8]=5.5+8.0j$$

$$X[3]=X^*[(9-3)_9]=X^*[6]=9.3-6.3j$$

$$X[5]=X^*[(9-5)_9]=X^*[4]=-1.7-5.2j$$

$$X[7]=X^*[(9-7)_9]=X^*[2]=2.5-4-6j$$

Properties of the Discrete Fourier Transform

(4) Circular convolution

$$y[k] = x_1[k] \textcircled{N} x_2[k] = \sum_{n=0}^{N-1} x_1[n] x_2[(k-n)_N]$$

Linear Convolution $$y[k] = x_1[k] * x_2[k] = \sum_{n=-\infty}^{\infty} x_1[n] x_2[k-n]$$

Circular Convolution $$\tilde{y}[k] = \tilde{x}_1[k] * \tilde{x}_2[k] = \sum_{n=0}^{N-1} \tilde{x}_1[n] \tilde{x}_2[k-n]$$

Properties of the Discrete Fourier Transform

(4) Circular convolution

$$y[k] = x_1[k] \textcircled{N} x_2[k] = \sum_{n=0}^{N-1} x_1[n] x_2[(k-n)_N]$$

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \end{bmatrix} = \begin{bmatrix} x_2[0] & x_2[3] & x_2[2] & x_2[1] \\ x_2[1] & x_2[0] & x_2[3] & x_2[2] \\ x_2[2] & x_2[1] & x_2[0] & x_2[3] \\ x_2[3] & x_2[2] & x_2[1] & x_2[0] \end{bmatrix} \begin{bmatrix} x_1[0] \\ x_1[1] \\ x_1[2] \\ x_1[3] \end{bmatrix}$$

Properties of the Discrete Fourier Transform

(4) Circular Convolution Theorem

※ Time-domain Convolution Theorem:

$$x_1[k] \textcircled{N} x_2[k] \xleftrightarrow{\text{DFT}} X_1[m]X_2[m]$$

※ Frequency-domain Convolution Theorem :

$$x_1[k]x_2[k] \xleftrightarrow{\text{DFT}} \frac{1}{N} X_1[m] \textcircled{N} X_2[m]$$

The cyclic convolution in the time domain corresponds to the product in the frequency domain.

The product in the time domain corresponds to the cyclic convolution in the frequency domain.

Example: Given $x[k]=\{1, 2, 3, 4\}$, $h[k]=\{5, 6, 7\}$, find $x[k]$, $h[k]$

(1)Circular convolution at the points 4, 5, and 6 (2)Linear convolution

Solution:

4points

$$y[k]=\begin{bmatrix}1&4&3&2\\2&1&4&3\\3&2&1&4\\4&3&2&1\end{bmatrix}\begin{bmatrix}5\\6\\7\\0\end{bmatrix}=\begin{bmatrix}50\\44\\34\\52\end{bmatrix}$$

5points

$$y[k]=\begin{bmatrix}1&0&4&3&2\\2&1&0&4&3\\3&2&1&0&4\\4&3&2&1&0\\0&4&3&2&1\end{bmatrix}\begin{bmatrix}5\\6\\7\\0\\0\end{bmatrix}=\begin{bmatrix}33\\16\\34\\52\\45\end{bmatrix}$$

6points

$$y[k]=\begin{bmatrix}1&0&0&4&3&2\\2&1&0&0&4&3\\3&2&1&0&0&4\\4&3&2&1&0&0\\0&4&3&2&1&0\\0&0&4&3&2&1\end{bmatrix}\begin{bmatrix}5\\6\\7\\0\\0\\0\end{bmatrix}=\begin{bmatrix}5\\16\\34\\52\\45\\28\end{bmatrix}$$

$x[k]*h[k]$

		$x[0]$	$x[1]$	$x[2]$	$x[3]$
		1	2	3	4
$h[0]$	5	5	10	15	20
$h[1]$	6	6	12	18	24
$h[2]$	7	7	14	21	28

$$=\begin{bmatrix}5\\16\\34\\52\\45\\28\end{bmatrix}$$

Example: Given $x[k]=\{1, 2, 3, 4\}$, $h[k]=\{5, 6, 7\}$, find $x[k]$, $h[k]$

(1)Circular convolution at the points 4, 5, and 6 (2)Linear convolution

Solution:

$$y[k]=\begin{bmatrix}1&0&0&4&3&2\\2&1&0&0&4&3\\3&2&1&0&0&4\\4&3&2&1&0&0\\0&4&3&2&1&0\\0&0&4&3&2&1\end{bmatrix}\begin{bmatrix}5\\6\\7\\0\\0\\0\end{bmatrix}=\begin{bmatrix}5\\16\\34\\52\\45\\28\end{bmatrix}\qquad x[k]*h[k]=\begin{bmatrix}5\\16\\34\\52\\45\\28\end{bmatrix}$$

The results of 6-point circular convolution and linear convolution are the same.

$$x_1[k]\textcircled{L}h_1[k]=x[k]*h[k]\qquad L=4+3-1=6$$

Properties of the Discrete Fourier Transform

If the length of $x[k]$ is N, and the length of $h[k]$ is M, then the circular convolution of $L \geq N+M-1$ is equal to the linear convolution of $x[k]$ and $h[k]$.

$$x_1[k] \textcircled{L} h_1[k] = x[k] * h[k]$$

Using the circular convolution property of DFT, linear convolution can be calculated through DFT.

$$x_1[k] \textcircled{L} h_1[k] \xleftrightarrow{\text{DFT}} X_1[m]H_1[m]$$

Given $x[k]=\{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$, Without calculating DFT, try to determine

(1) $X[0]$ (2) $X[3]$ (3) $\sum_{m=0}^{5} X[m]$ (4) $\sum_{m=0}^{5} \mathrm{e}^{-\mathrm{j}(\pi m/3)} X[m]$ (5) $\sum_{m=0}^{5} \left|X[m]\right|^2$

Solution: $X[m]=\sum_{m=0}^{N-1} x[k]\cdot \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}mk}$ $x[k]=\frac{1}{N}\sum_{m=0}^{N-1} X[m]\cdot \mathrm{e}^{\mathrm{j}\frac{2\pi}{N}mk}$

(1) $X[0]=\sum_{k=0}^{5} x[k]$ =21

(2) $X[3]=\sum_{k=0}^{5} x[k]\mathrm{e}^{-\mathrm{j}\frac{2\pi}{6}3k}=\sum_{k=0}^{5} x[k]\mathrm{e}^{-\mathrm{j}k\pi}=\sum_{k=0}^{5} (-1)^k x[k]$ =-3

(3) $\sum_{m=0}^{5} X[m]=6x[0]$ =6

Given $x[k]=\{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$, Without calculating DFT, try to determine

(1) $X[0]$ (2) $X[3]$ (3) $\sum_{m=0}^{5} X[m]$ (4) $\sum_{m=0}^{5} e^{-j(\pi m/3)} X[m]$ (5) $\sum_{m=0}^{5} |X[m]|^2$

Solutioin: (4) $x[(k-n)_N] \xleftrightarrow{\text{DFT}} e^{-j\frac{2\pi}{N}mn} X[m]$ $x[k]=\frac{1}{N}\sum_{m=0}^{N-1} X[m]\cdot e^{j\frac{2\pi}{N}mk}$

$$e^{-j(\pi m/3)} X[m] = e^{-j\frac{2\pi}{6}m} X[m] \longrightarrow x[(k-1)_6]$$

$$\sum_{m=0}^{5} e^{-j(\pi m/3)} X[m] = 6x[(0-1)_6] = 6\cdot x[5] = 36$$

(5) $$\sum_{m=0}^{5} |X[m]|^2 = 6\sum_{k=0}^{5} |x[k]|^2 = 546$$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude